Integrated Energy-Economy Model

The file “Model_IEEM” in Excel file implements the model described below. It has 5 sheets. Two for each scenario: Business as Usual (BAU) and Decarbonization and one with charts.

For each scenario:

- Fill in the “Aggregated_efficiency_BAU” sheet: Fill in the cells in yellow. This sheet estimates the aggregated final to useful efficiency up to 2050.
- Fill in the “BAU” sheet: Fill in the cells in yellow. This sheet estimates the GHG emissions, GDP and final energy.
- Fill in the “Aggregated_efficiency_Decarbonization” sheet: Fill in the cells in yellow. This sheet estimates the aggregated final to useful efficiency up to 2050.
- Fill in the “Decarbonization” sheet: Fill in the cells in yellow. This sheet estimates the GHG emissions, GDP and final energy.

Description of the Model

This section is based on work developed in the context of the project (Maffia, 2025; Sousa et al., 2026) building on preliminary work (Alvarenga et al., 2017). 2.1 Brief description of the model

The model operates on annual time steps, integrating demographic, economic, and energy-exergy dynamics within a unified framework. It comprises four interconnected components: (1) the core module linking energy and the economy – see previous section; (2) the human labour module; (3) the capital module; and (4) the energy system module. The conceptual structure is illustrated in Figure 1 and implemented in Excel. Scenario development requires scenarios for the variables highlighted in red in ***Figure 1***.

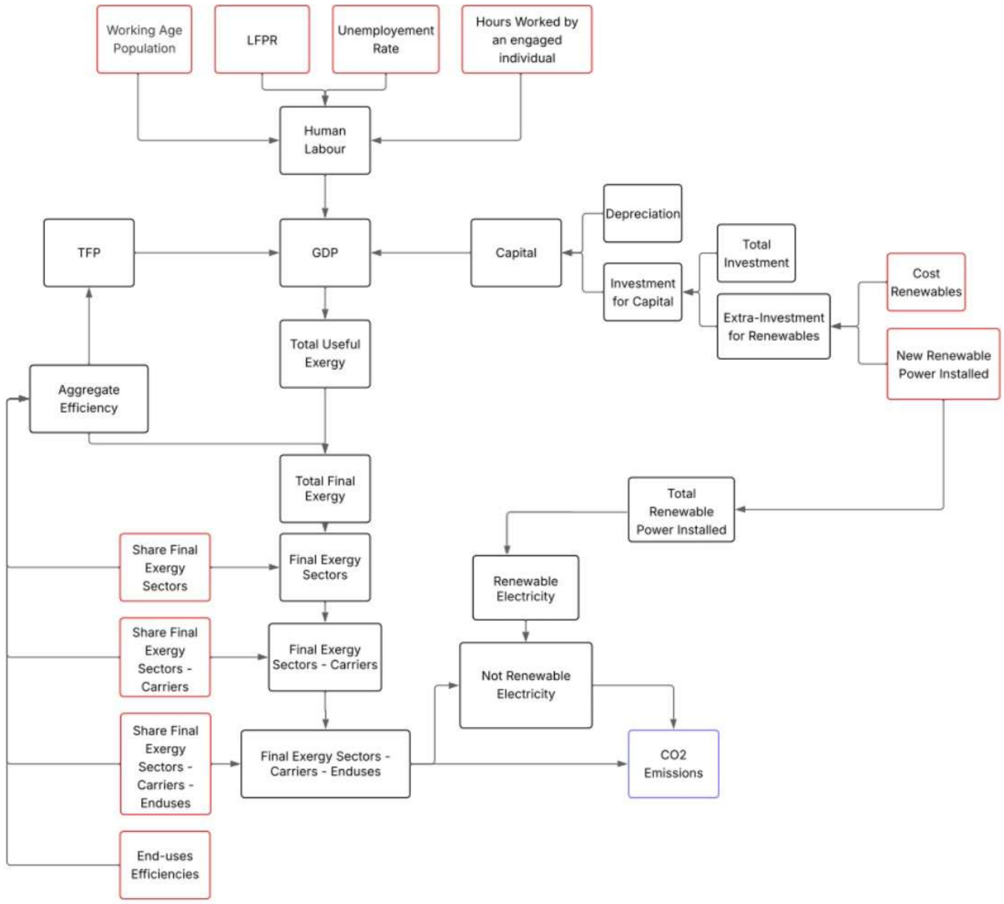

Figure 1 - Conceptual map of the model where inputs are shown in red (Maffia, 2025).

Energy-economy module

Economic output is represented using a Cobb-Douglas production function:

$$Y = A \cdot K^{\alpha_K} \cdot L^{\alpha_L}$$

where α_L is typically set to 0.3 and α_K to 0.7. Here, L represents human labour, measured as the total number of hours worked by all employed individuals in the economy, as estimated in the human labour module. K denotes the capital inputs estimated in the capital module. A is the total factor productivity.

For Portugal, total factor productivity A is estimated as a function of the aggregate final-to-useful efficiency of the economy through the following empirical relationship:

$$A_t = \left(\frac{\varepsilon_t}{\varepsilon_{1960}}\right)^{1.0480}$$

This relationship is country-specific and can be derived within the Efficiency-Total Factor Productivity model developed in this project (previous section). Rather than treating TFP growth as exogenous, the model assumes a direct relationship between TFP and aggregate efficiency. By defining TFP as a function of the efficiency with which final exergy is converted into useful exergy – a proxy for technological progress - the model provides a non-arbitrary and thermodynamically consistent representation of TFP. Aggregate efficiency is calculated annually in the energy module, based on the allocation of exergy across sectors, energy carriers, and end uses, along with their respective efficiencies.

For Portugal, total useful exergy demand is assumed to be directly proportional to GDP. Empirically, 1 billion euros of GDP corresponds to approximately 1000 TJ of total useful exergy, based on historical data for the period 1859-2009:

$$X_{useful} = GDP_t \cdot 10^3$$

This relationship is country-specific and can be derived within the Efficiency-Total Factor Productivity model (previous section). If historical useful exergy intensity is deemed constant by the kit detailed in Section 1, then the mean over the whole sample is used in linking future projections of GDP to future projections of total useful exergy consumed. If, however, historical useful exergy intensity is not deemed constant, choices are: 1) log-linear trend extrapolation from the whole sample; 2) mean of last BIC-detected segment (if approximately constant); 3) log-linear extrapolation of last BIC-detected segment.

Human labour module

Human labour measures the total number of hours worked by all engaged individuals in the economy, integrating demographic and labour market characteristics:

$$L_t = WAP_t \cdot LFPR_t \cdot (1 - u_t) \cdot H_t$$

Working-age population (WAP) refers to the segment of the population typically considered of working age, between 15 and 64 years. The labour force participation rate ($LFPR$) measures the share of this population that is active in the labour market, either employed or actively seeking employment. The unemployment rate (u) represents the proportion of the labour force that is without a job but actively looking for one. H denotes the average number of hours worked per employed individual.

Capital module

Capital, along with human labour, is a major factor in economic output in the model. Annual changes in capital stock depend on investment and depreciation:

$$K_{s,t} = K_{s,t-1} + I_{s,t} - D_{s,t}$$

To increase accuracy, capital (K), investment (I) and depreciation (D) are divided into sectors (s). Thus, the model can reflect the characteristics of each sector and show how capital accumulation is linked to the historical economic structure of the region.

The total capital stock is the sum across all sectors:

$$K_t = \sum_s K_{s,t}$$

The subscript s indicates the different sectors considered including transport, industry, residential, services and agriculture/fisheries/forestry.

To capture the rapid expansion of the renewable energy sector in the Decarbonization scenario—beyond the levels observed in the BAU scenario—the cost of additional renewable capacity (i.e., capacity installed above BAU levels) is subtracted from total investment. This reflects the fact that this portion of investment is effectively imposed to meet the roadmap targets and would otherwise be available for other types of investment. To estimate the cost of the extra renewable capacity, the prices of each type of renewable energy and the new capacities are required.

For each renewable source i:

$$RC_{extra,t,i} = RC_{DEC,t,i} - RC_{BAU,t,i}$$

The total extra investment is:

$$I_{extra,t} = \sum_i RC_{extra,t,i} C_{t,i}^{GW}$$

where $C_{t,i}^{GW}$ represents the installation cost per GW of renewable energy.

This additional investment is then subtracted from standard investment:

$$I_{adj,s,t} = I_{s,t} - I_{extra,t}$$

This adjustment maintains realism in capital reallocation for accelerated energy transitions making the energy transition financially explicit.

Energy system module

Total useful exergy is converted into total final exergy using aggregate efficiency:

$$X_{final,t} = \frac{X_{useful,t}}{\varepsilon_t}$$

The total final exergy is distributed between economic sectors (industry, transport, residential, services, agriculture/forestry/fishing, and energy own use), energy carriers (coal, oil, gas, renewables, electricity, waste, etc.), and finally end uses (e.g., high-temperature heat, low-temperature heat, mechanical drive, etc.) using the defined allocations.

The final exergy allocated to each sector s is:

$$X_{final,s,t} = SX_{final,s,t} \cdot X_{final,t}$$

where $SX_{\mathrm{final,s}}$ is the share of final exergy allocated to sector s.

For each sector s, the final exergy allocated to each energy carrier c is:

$$X_{final,s,c,t} = SX_{final,s,c,t} \cdot X_{final,s,t}$$

For each sector-energy carrier pair, the final exergy allocated to each end use e is:

$$X_{\mathrm{final,s,c,e,t}} = SX_{\mathrm{final,s,c,e,t}} \cdot X_{\mathrm{final,s,c,t}}$$

Aggregate efficiency is the ratio between useful exergy and final exergy. Since the shares and end-use efficiencies are known a priori, it is possible to compute aggregate efficiency for the whole series:

$$\varepsilon_t = \sum_s SX_{final,s,t} \left(\sum_c SX_{final,s,c,t} \left(\sum_e SX_{final,s,c,e,t} \cdot \varepsilon_{c,e,t} \right) \right)$$

These efficiencies and allocations allow the annual aggregate exergy efficiency to be calculated, which in turn feeds back into the calculation of TFP, GDP and useful exergy.

Having final exergy by carrier from the complete breakdown, the model derives CO2 emissions by multiplying emission factors (EF), measured in tonnes of CO2 per TJ, by the final exergy use of each carrier:

$$CO2_{c,t} = EF_c \cdot X_{final,s,t}$$

If renewable electricity supply is insufficient to meet electricity demand, the model assumes that natural gas is used to compensate. Renewable electricity, *RX*, from each source *i* (wind, solar, water) is computed as:

$$RX_{electricity,i,t} = RC_{i,t} \cdot CF_{i,t} \cdot 8740$$

The capacity factor (*CF*) measures how much an energy source produces relative to maximum theoretical capacity. It is defined as the ratio between actual energy generation and the output that would result from operating the installed capacity (*RC*) at full power for all hours of the year, where 8760 is the total number of hours in a year. *CF* thus reflects resource availability and operational constraints, depends on the type of renewable technology (e.g., wind, solar), and is key for translating installed capacity into actual energy production. If:

$$X_{electricity,t} > RX_{electricity,t}$$

natural gas is used to produce the remaining electricity demand and additional emissions, $\mathrm{CO2}_{\mathrm{electricity}}$ are produced. These emissions are added to the other emissions to obtain the total:

$$CO2_t = CO2_{electricity,t} + \sum_c CO2_{c,t}$$

The final indicator produced by the model is carbon intensity:

$$CI_t = \frac{CO2_t}{GDP_t}$$

This metric evaluates the degree of decoupling between economic activity and environmental impact. A declining carbon intensity indicates a cleaner growth trajectory.

Data and indicators

The IEEM requires annual demographic, macroeconomic, capital, energy and emissions data. These include working-age population, labour force participation, unemployment, hours worked, sectoral capital stocks, investment and depreciation, sectoral allocations of final exergy by carrier and end use, end-use efficiencies, renewable capacities, capacity factors, and emission factors. The main indicators produced by the model are GDP, TFP, human labour, capital stock, useful exergy, final exergy, aggregate final-to-useful exergy efficiency, CO2 emissions and carbon intensity.

Input data:

Working age population (WAP) represents the part of the population that is generally considered suitable for work, and it varies from 15 to 64 years.

For Portugal:

The BaU scenario follows the central projection of the National Statistics Institute (INE, 2020), with the WAP declining from 6.5 million to 5.5 million. For the Decarbonization Scenario, the roadmap is improved with more recent data from the INE. According to this projection, the WAP decreases from 6.5 million to approximately 6 million for the same time.

Instituto Nacional de Estatística. *Portal do Instituto Nacional de Estatística*. Retrieved 2025, from https://www.ine.pt/xportal/xmain?xpgid=ine_main&xpid=INE

Labour Force Participation Rate (LFPR) indicates the share of the working age population who are currently engaged in the labour market, either employed or actively seeking for work.

For Portugal:

The historical data used are provided by the International Labour Organization, using the ILOSTAT data explorer, and by the INE database up to 2024. In the BaU scenario, it increases from 0.792, last data available, up to 0.85 in 2050. In the Decarbonization Scenario, it reaches 0.9, assuming inclusive polices and structural improvements in workforce participation.

International Labour Organization. ILOSTAT database. Retrieved 2025, from https://ilostat.ilo.org/

Unemployment rate (u) measures the percentage of the active workforce that does not have a job but is searching for it.

For Portugal:

Historical data comes from the INE database up to 2024 and in the BaU scenario this term is assumed to maintain the same value as it was in 2024, i.e. 6.4%. Instead, in the Decarbonization Scenario the unemployment rate will decrease up to 5 % by 2050, the 2023 record global average stated by (United Nations, 2024).

United Nations. (2024). The sustainable development goals report 2024. United Nations. https://unstats.un.org/sdgs/report/2024/The-Sustainable-Development-Goals-Report-2024.pdf

Average Hours Worked by an engaged individual (H) gives an idea of the labour intensity per worker.

For Portugal:

Historical data from Penn World Table database outlines the baseline up to 2018 and then for the BaU scenario the last historical year value is kept constant, i.e.1864.5 hours worked per year, while for the Decarbonization Scenario it gradually decreases up to 1450 hours per year in 2050, resulting from 8 hours of work per day, 4 days of work per week and 45.3 weeks of work per year, showing a transition to a 4-day workweek.

Penn World Table. (2023). Penn World Table version 10.01. Groningen Growth and Development Centre, University of Groningen. https://www.rug.nl/ggdc/productivity/pwt/

All historical data for **Capital, Investments and Depreciation** are obtained from the Eurostat database.

For Portugal:

For the Business-as-usual scenario, the yearly investment in a sector is calculated by the product of the average investment-to-GDP ratio, historically observed from 2000 to 2022, and the GDP of the previous year. While for the Decarbonization scenario, the Portuguese roadmap to carbon neutrality – RNC2050, assumptions are followed, since it gives yearly investments divided by sectors. To get the yearly depreciation in a sector, the average of depreciation-to-capital ratio, which is also recorded from 2000 to 2022, is applied to the previous year's capital stock for both the scenarios.

Eurostat. Eurostat database. Retrieved 2025, from https://ec.europa.eu/eurostat

Costs of new renewable capacity are based on historical data from (IRENA, 2024).

For Portugal:

Costs are assumed to continue declining along recent trends. Specifically, costs are modelled to decrease exponentially until reaching a lower bound, after which they remain constant. For hydrogen, future cost projections are taken directly from (IRENA, 2020). These assumptions are applied consistently across both scenarios.

International Renewable Energy Agency. (2020). Green hydrogen cost reduction: Scaling up electrolysers to meet the 1.5°C climate goal. https://www.irena.org/publications/2020/Dec/Green-hydrogen-cost-reduction

International Renewable Energy Agency. (2024). Renewable power generation costs in 2023. https://www.irena.org/Publications/2024/Aug/Renewable-power-generation-costs-in-2023

Historical data for **installed renewable capacity** are derived from (IRENA, 2025).

For Portugal:

In the BaU scenario, the growth in installed renewable capacity follows historical trends. In contrast, the Decarbonization scenario aligns with the RNC2050 targets, implying a strong acceleration in the deployment of solar, wind, and other renewable technologies, as well as the introduction of hydrogen.

International Renewable Energy Agency. (2025). Renewable capacity statistics 2025. https://www.irena.org/Publications/2025/Mar/Renewable-Capacity-Statistics-2025

Historical information for the **final exergy allocated to each sector, the final exergy of each sector allocated to each energy carrier and the final exergy of each pair sector-energy carrier allocated to each end use** are derived from IEA statistics.

For Portugal:

In the BaU scenario sectoral exergy shares of final exergy follow historical trends 1960-2022, relying on their long-term linearity, carrier shares and end-use distributions are obtained for 2012-2022, since post-crisis restructuring. For the Decarbonization Scenario, all the shares follow RNC2050 projections and assumptions. Individual carrier-end-use efficiencies for both scenarios are extrapolated from their historical trends 1960-2022.